



VRIJE
UNIVERSITEIT
BRUSSEL



Graduation thesis submitted in partial fulfilment of the requirements for the
degree of de Ingenieurswetenschappen: Computerwetenschappen

PROGRAMMING ROBOTS WITH HIGHER-ORDER REACTIVE PROGRAMMING

Gérard Lichtert

2025-2026

Promotor: Prof. Dr. Wolfgang De Meuter
Advisor: Bjarno Oeyen

Sciences and Bio-engineering Sciences



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Proefschrift ingediend met het oog op het behalen van de graad van Master of
Science in de Ingenieurswetenschappen: Computerwetenschappen

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Wetenschappen en Bio-ingenieurswetenschappen

Abstract

Your abstract would go here.

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Chapter 1

Introduction

In today's world, avoiding electronics or their applications is unthinkable.

1.1 Contributions

1.2 Overview of this Dissertation

1.3 Approach

Chapter 2

State of the Art

2.1 Reactive Programming

2.2 RP languages targeting MCUs

2.3 Systems to program robots

2.4 Tierless Programming

2.5 Problem Statement

Chapter 3

μ -Remus

3.1 Haai & Remus

3.2 ROS & μ -ROS

3.3 Mapping RP to ROS & μ -ROS

3.4 μ -Remus: A C implementation

Chapter 4

Programming Robots with Reactors

4.1 Linking ROS and μ -ROS

Chapter 5

Tierless Haai

5.1 Haai to CHaai

Chapter 6

Evaluation

6.1 Evaluating HaaI distribution

6.2 Evaluating HaaI to CHaaI

6.3 Evaluating CHaaI on μ -Remus

Chapter 7

Conclusion & Future Work

